Table 1: T&O: Descriptives

Panel A: Organizational changes and alternative measures of technological change			
	Number of changes introduced over past 3 years		
	0	1 or 2	3 or 4
(i) percentage of firms	44.32%	43.29%	12.39%
(ii) # investments in IT (sum of yearly dummies)	1.31	1.73	1.79
p-value		0.00	0.00
(iii) investment in IT per employee (=1 in firms with no change)		1.63	2.38
p-value		0.00	0.00
(iv) dummy for product or process innovations (3 year period)	0.11	0.20	0.29
p-value		0.00	0.00
Panel B: Number of organizational changes and firm characteristics (at baseline)			
	Number of changes introduced over past 3 years		
	0	1 or 2	3 or 4
(i) firm size	44.16	89.92	134.27
p-value		0.00	0.00
(ii) firm daily wage (in logs)	4.40	4.49	4.49
p-value		0.00	0.00
(iii) fixed firm effect (relative to nonrestructuring firms)		2.68%	2.45%
p-value		0.00	0.00
(iv) fixed worker effect (relative to nonrestructuring firms)		2.93%	1.33%
p-value		0.00	0.02
(v) baseline task usage, in percentages			
abstract	39.95%	46.06%	43.89%
p-value		0.00	0.00
routine	39.59%	38.08%	40.24%
p-value		0.14	0.55
manual	20.46%	15.86%	15.87%
p-value		0.00	0.00
(vi) baseline age structure, in percentages			
<30	19.83%	19.64%	21.04%
p-value		0.49	0.00
31-50	57.36%	58.04%	57.38%
p-value		0.01	0.96
>50	22.81%	22.32%	21.58%
p-value		0.04	0.00

Notes: The table reports correlations between the number of organizational changes (T&O) and alternative measures of technological change (Panel A) and firm characteristics measured at baseline, before organizational change takes place (Panel B). To construct the number of organizational changes, we sum information from four binary variables for whether firms transfer responsibilities to subordinates, introduce team work or self-responsible working groups, introduce profit centers, and/or pool or restructure internal departments. The first row of panel A reports the share of firms by level of organizational change. The average number of organizational changes across all years is 1.01 (1.81 conditional on at least one change). In panel A, the value of IT investments per employee—calculated from total investment in euros, share of IT investment in total investment, and employment levels—is normalized to one for firms that implement no organizational change. In panel B, fixed firm effects and fixed worker effects are pre-estimated from social security data using the previous seven calendar years for each point in time. All statistics are weighted by employment to make them representative of workers (larger firm = larger weight) and use firm panel weights to correct for oversampling of larger firms and certain industries in the IAB Establishment Panel. In all panels, the p-values are derived from pseudo-regressions in which standard errors are clustered at the firm level.

Sources: Registry data and the IAB Establishment Panel, 1993–2010. We calculate task shares based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

Table 2: Differences between Manual, Routine, and Abstract Tasks

	routine	manual	abstract
employment shares	37.11%	17.70%	45.19%
Panel A			
mean routine task index	0.594	0.247	0.209
mean manual task index	0.194	0.569	0.091
mean abstract task index	0.212	0.184	0.701
Panel B			
mean wage (percentage difference relative to routine)		2.63%	33.41%
Percentage low-edu	18.99%	6.34%	3.09%
Percentage medium-edu	77.99%	90.99%	66.83%
Percentage high-edu	3.03%	2.68%	30.08%
Percentage ages <30	17.91%	18.25%	15.68%
Percentage aged 30-49	57.04%	56.90%	59.87%
Percentage aged 50+	21.89%	21.48%	21.79%

Notes: The first row of the table displays the share of workers in each task in our sample. Panel A shows the task indices, continuous measures of the prevalence of each task (see Section 2.4), across workers in each of the three task groups. The first row of panel B compares average wages in the three task groups, using imputed log wages derived as detailed in the Appendix (cf. Dustmann, Ludsteck, and Schönberg, 2009; Card, Heining, and Kline, 2013). The other rows of panel B display education and age employment shares in the three task groups (e.g., the share of workers with intermediate education among routine workers). All statistics are weighted by employment to make them representative of workers (larger firm = larger weight) and use firm panel weights to correct for oversampling of larger firms and certain industries in the IAB
Sources: Registry data and the IAB Establishment Panel, 1993–2010. We calculate task shares based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

Table 3: T&O and the Firm's Task, Education, and Age Structure

Panel A: Tasks

	Occupational Tasks					
	Manual		Routine		Abstract	
	(1)	(2)	(1)	(2)	(1)	(2)
number of changes	0.015	0.029	-0.363***	-0.227**	0.347**	0.198
	(0.054)	(0.060)	(0.122)	(0.111)	(0.151)	(0.145)
year fixed effects	yes	yes	yes	yes	yes	yes
firm size, commuting area, industry	no	yes	no	yes	no	yes
mean change in employment share	-0.308%		-1.222%		1.530%	

Panel B: Education

	Educational Groups					
	Low		Medium		High	
	(1)	(2)	(1)	(2)	(1)	(2)
number of changes	-0.043	-0.008	-0.249*	-0.131	0.291**	0.139
	(0.057)	(0.041)	(0.144)	(0.120)	(0.128)	(0.105)
year fixed effects	yes	yes	yes	yes	yes	yes
firm size, commuting area, industry	no	yes	no	yes	no	yes
mean change in employment share	-0.944%		0.014%		0.930%	

Panel C: Age

	Age Groups					
	<30		30-49		50-65	
	(1)	(2)	(1)	(2)	(1)	(2)
number of changes	-0.033	-0.043	0.370***	0.308**	-0.337**	-0.265*
	(0.065)	(0.073)	(0.115)	(0.135)	(0.131)	(0.148)
year fixed effects	yes	yes	yes	yes	yes	yes
firm size, commuting area, industry	no	yes	no	yes	no	yes
mean change in employment share	-3.253%		1.547%		1.707%	
number of observations	26,908	24,939	26,908	24,939	26,908	24,939

Panel D: Placebo from Tasks Regressions

	Occupational Tasks					
	Manual		Routine		Abstract	
	(1)	(2)	(1)	(2)	(1)	(2)
contemporenuous number of changes	-0.026	-0.029	-0.352***	-0.227**	0.378***	0.257**
	(0.043)	(0.046)	(0.082)	(0.089)	(0.090)	(0.103)
future number of changes	0.071*	0.089*	-0.042	0.027	-0.029	-0.116
	(0.041)	(0.047)	(0.102)	(0.105)	(0.110)	(0.117)
year fixed effects	yes	yes	yes	yes	yes	yes
firm size, commuting area, industry	no	yes	no	yes	no	yes
number of observations	12,450	11,893	12,450	11,893	12,450	11,893

Notes: The table reports estimates for the effects of T&O on the task (Panel A), education (Panel B), and age (Panel C) structure of the firm. Panel D investigates the same groups (tasks) as in Panel A, but includes in the regressions the number of future organizational changes (implemented between t and t+3) in addition to the number of contemporaneous organizational changes (carried out between t-3 and t). The dependent variable in each panel and column is the change in the firm level employment share of the specific group (e.g., routine workers) between t-3 and t. The main variable of interest, "number of changes," which measures the number of organizational changes between t-3 and t, ranges between zero and four. Employment shares vary from 0 to 100; hence, the coefficients reported in the table refer to the effect of one additional organizational change on the percentage point change in the respective employment share. In the first set of column, we control only for year fixed effects (as in equation (1) in the text); in the second, we add in controls (as in equation (2) in the text) for firm size (measured as log employment at baseline), commuting zones (a regional control derived by dividing West Germany into 50 regions based on commuting patterns), and industry (one digit controls). The unit of observation is the firm/year, and all regressions are weighted by (baseline) employment. Standard errors are clustered at the firm level; the significance levels are *10%, **5%, ***1%. In the last rows of each panel, the table shows the mean changes in employment shares of the specific group. To compute these, we again weight by (baseline) employment and additionally use firm panel weights to correct for oversampling of larger firms and certain industries in the IAB Establishment Panel.

Sources: Registry data and the IAB Establishment Panel, 1993–2010. We calculate task shares based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

Table 4: T&O, Wage and Employment Growth, Churning, and Internal Reshuffling

	Wage growth	Employment growth	External separations	External hiring	Internal reshuffling
number of changes	0.074	0.473	-0.199	0.274**	0.131**
	(0.123)	(0.328)	(0.316)	(0.117)	(0.053)
number of observations	25821	25866	25866	25866	25866
mean	7.637%	-8.284%	11.605%	19.889%	1.198%

Notes: Notes: The table reports estimates for the effects of T&O on firm wage growth (the log difference in mean wages in the firm between t-3 and t); employment growth (employment in t minus firm employment in t-3, divided by firm employment in t-3); external hiring (the number of workers who joined the firm between t-3 and t, divided by firm employment in t-3); external separations (the number of workers who left the firm over the 3-year period between t-3 and t, divided by employment at t-3); and internal reshuffling (the number of individuals who have changed task within the firm over this 3-year period, divided by baseline employment). The main variable of interest, "number of changes," which measures the number of organizational changes between t-3 and t, ranges between zero and four. All regressions control (as in equation (2) in the text) for firm size (measured as log employment at baseline), commuting zones (a regional control derived by dividing West Germany into 50 regions based on commuting patterns), and industry (one digit controls). The unit of observation is the firm/year, and all regressions are weighted by (baseline) employment. Standard errors are clustered at the firm level; the significance levels are *10%, **5%, ***1%. The last row of the table shows the mean of each dependent variable in our sample. To compute these, we again weight by (baseline) employment and additionally use firm panel weights to correct for oversampling of larger firms and certain industries in the IAB Establishment Panel.

Sources: Registry data and IAB Establishment Panel, 1993–2010. We calculate tasks for the internal reshuffling regression based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

Table 5: T&O and Career Outcomes, by Tasks and Age at Baseline

	Working	Working at the same task	Working at a different task, but at the same firm	Working at the same firm	Wage growth
	(1)	(2)	(3)	(4)	(5)
Panel A: By tasks					
(1) Manual					
number of organizational changes	0.034	-0.026	0.120	0.709**	0.140*
	(0.135)	(0.185)	(0.116)	(0.335)	(0.075)
mean of dependent variable	83.238%	77.021%	1.974%	67.792%	6.000%
(2) Routine					
number of organizational changes	-0.023	-0.242*	0.195***	0.528	0.141*
	(0.123)	(0.144)	(0.072)	(0.367)	(0.082)
mean of dependent variable	82.210%	77.277%	1.608%	65.665%	6.806%
(3) Abstract					
number of organizational changes	-0.078	-0.093	0.043**	0.648	-0.062
	(0.100)	(0.108)	(0.021)	(0.406)	(0.084)
mean of dependent variable	83.190%	81.119%	0.521%	65.534%	9.807%
Panel B: By age					
(1) <30					
number of organizational changes	0.037	-0.175	0.205**	0.540	0.122
	(0.074)	(0.129)	(0.088)	(0.337)	(0.111)
mean of dependent variable	81.800%	74.933%	1.512%	58.074%	13.308%
(2) 30-49					
number of organizational changes	0.134*	0.029	0.141**	0.872**	0.067
	(0.074)	(0.097)	(0.055)	(0.366)	(0.079)
mean of dependent variable	89.875%	85.922%	1.292%	71.929%	7.345%
(5) >49					
number of organizational changes	-0.649*	-0.699**	0.051*	-0.205	0.028
	(0.333)	(0.330)	(0.030)	(0.397)	(0.063)
mean of dependent variable	65.394%	63.888%	0.724%	56.897%	3.977%
Panel C: Detailed older age groups					
(1) 50-54					
number of organizational changes	-0.231	-0.312	0.087**	0.277	0.028
	(0.425)	(0.424)	(0.041)	(0.538)	(0.067)
mean of dependent variable	80.008%	77.942%	0.935%	68.466%	4.316%
(2) 55-59					
number of organizational changes	-1.362***	-1.371***	0.003	-0.966***	0.046
	(0.353)	(0.345)	(0.020)	(0.338)	(0.069)
mean of dependent variable	53.123%	52.165%	0.535%	47.664%	3.366%
(3) >59					
number of organizational changes	-0.592**	-0.591**	0.001	-0.386*	-0.103
	(0.235)	(0.233)	(0.014)	(0.210)	(0.144)
mean of dependent variable	24.714%	24.412%	0.189%	22.530%	2.754%

Notes: The table reports estimates of the impact of T&O on the career paths over a three-year period of workers who were employed in one of the firms in the IAB Establishment Panel at baseline (in t-3), and were at most 62 at baseline; see equation (3) in the text. Panel A distinguishes between workers who performed manual, routine, or abstract tasks at baseline; Panels B and C distinguish between workers of different age groups (under 30, between 30 and 49, 50 and above at baseline in Panel B and between 50 and 54, between 55 and 59 and between 59 and 62 at baseline in Panel C). The main variable of interest, "number of changes," which measures the number of organizational changes in the baseline firm between t-3 and t, ranges between zero and four. We consider the following outcome variables: whether the worker is still employed at any firm in Germany at time t ("working"); whether the worker is employed in the same task at time t as at baseline ("working at the same task"); whether the worker is employed at a different task, but same firm, at time t ("working at a different task but at the same firm"); whether the worker is employed at the same firm irrespective of task at time t as at baseline ("working at the same firm"); and "wage growth", computed from imputed wages to deal with censoring (see Appendix), between t-3 and t. Whereas regressions in the first four columns refer to all workers who were employed in one of the firms in the IAB Establishment Panel at baseline—irrespective of whether or not they are employed at time t—the wage growth regressions in the last column are restricted to workers who are employed at time t. In the first four columns, the dependent variables take the values of either 0 or 100 so that the coefficients reported in the table refer to the impact of one additional organizational change on the percentage point change of the particular event. All regressions control for age and age squared at baseline, foreign status, year fixed effects, firm size (measured as log employment at baseline), commuting zones (a regional control derived by dividing West Germany into 50 regions based on commuting patterns), and industry (one digit controls). With the exception of the wage growth regressions, they also include pre-estimated fixed firm effects and fixed worker effects calculated using the seven years prior to the baseline year (cf. Card, Heining, and Kline, 2013). The unit of observation is the worker/year, and the sample includes 8,154,560 observations in the first four columns and 6,695,055 in the last (wage growth) column. Standard errors are clustered at the firm level; the significance levels are *10%, **5%, ***1%. The table also reports the means of the dependent variables. We compute these using the firm panel weights to correct for oversampling of larger firms and certain industries in the IAB Establishment Panel.

Sources: Registry data and IAB Establishment Panel, 1993–2010. We calculate tasks for the internal reshuffling regression based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

Table 6: T&O and Career Outcomes for Routine Workers

	Working	Manual task	Routine task	Abstract task	Abstract task, same firm	Abstract task, different firm
Panel A: Routine, all						
after 3 years	-0.023	0.038	-0.242*	0.183***	0.122***	0.060**
number of organizational changes	(0.123)	(0.062)	(0.144)	(0.047)	(0.038)	(0.025)
after 6 years	0.102	0.157*	-0.479**	0.220**	0.046	0.174***
number of organizational changes	(0.169)	(0.082)	(0.206)	(0.088)	(0.084)	(0.053)
Panel B: By age (after 3 years)						
<30	0.083	0.086	-0.233	0.231***	0.110*	0.121***
number of organizational changes	(0.090)	(0.092)	(0.151)	(0.077)	(0.058)	(0.043)
30-49	0.074	0.034	-0.145	0.185***	0.133***	0.052**
number of organizational changes	(0.099)	(0.070)	(0.136)	(0.052)	(0.043)	(0.026)
>49	-0.576	-0.018	-0.654	0.097***	0.081***	0.016
number of organizational changes	(0.404)	(0.029)	(0.399)	(0.032)	(0.027)	(0.017)
Panel C: Detailed age groups for workers older than 49 at baseline (after 3 years)						
50-54	-0.142	-0.028	-0.263	0.149***	0.120***	0.029
number of organizational changes	(0.535)	(0.038)	(0.534)	(0.044)	(0.035)	(0.026)
55-59	-1.314***	0.004	-1.341***	0.024	0.028	-0.004
number of organizational changes	(0.386)	(0.022)	(0.373)	(0.023)	(0.021)	(0.010)
>59	-0.514**	-0.041***	-0.485**	0.012	0.019	-0.007
number of organizational changes	(0.225)	(0.015)	(0.220)	(0.022)	(0.018)	(0.012)

Notes: The table reports estimates for the effects of T&O on career outcomes over a 3- and 6-year period of workers who were employed at a firm in the IAB Establishment Panel, performed routine tasks, and were at most 62 at baseline (in t-3); see equation (3) in the text. Panel A displays results pooled for all routine workers; Panels B and C further distinguish between different age groups (under 30, between 30 and 49, 50 and above at baseline in Panel B and between 50 and 54, between 55 and 59 and between 59 and 62 at baseline in Panel C). The main variable of interest in both the 3- and the 6-year regressions, “number of changes,” which measures the number of organizational changes in the baseline firm between t-3 and t, ranges between zero and four. We consider the following outcome variables: whether the individual is employed (at any firm in Germany) at time t in the 3-year regressions and at time t+3 in the 6-year regressions (“working”); whether the individual is employed in a manual, routine or abstract occupation at time t in the 3-year regressions or t+3 in the 6-year regressions (“manual task”, “routine task”, “abstract task”); and whether the worker is employed in an abstract occupation within the same or a different firm at time t or t+3 (“abstract task, same firm” and “abstract task, different firm”). The dependent variables take the values of either 0 or 100 so that the coefficients reported in the table refer to the impact of one additional organizational change on the percentage point change of the particular event. All regressions control for age and age squared at baseline, foreign status, year fixed effects, firm size (measured as log employment at baseline), commuting zones (a regional control derived by dividing West Germany into 50 regions based on commuting patterns), industry (one digit controls), pre-estimated fixed firm effects and fixed worker effects calculated using the seven years prior to the baseline year (cf. Card, Heining, and Kline, 2013). The unit of observation is the worker/year, and the full sample (all routine workers at baseline) includes 8,154,560 observations. Standard errors are clustered at the firm level; the significance levels are *10%, **5%, ***1%.

Sources: Registry data and IAB Establishment Panel, 1993–2010. We calculate tasks based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

Table 7: T&O and Training

	(1) All	(2) low-skilled	(3) medium-skilled	(4) high-skilled
mean employment share in training	17.61%	7.54%	18.41%	22.84%
number of organizational changes	1.278***	0.842*	2.513***	2.960***
	(0.321)	(0.454)	(0.388)	(0.475)
number of observations	17,199	12,788	15,212	12,786

Notes: The table reports estimates for the effects of T&O on firms' training activities. The main variable of interest, "number of organizational changes", which measures the number of organizational changes over three years between t-3 and t, ranges between zero and four. The dependent variables are the shares of workers (all workers in column (1) and low-, medium-, and high-skilled workers in columns (2) to (4)) who received further training between t-3 and t. These shares vary from 0 to 100; hence, the coefficients reported in the table refer to the effect of one additional organizational change on the percentage point change in the respective training share. All regressions control (as in equation (2) in the text) for year fixed effects, firm size (measured as log employment at baseline), commuting zones (a regional control derived by dividing West Germany into 50 regions based on commuting patterns), and industry (one digit controls). The unit of observation is the firm/year, and all regressions are weighted by (baseline) employment. Standard errors are clustered at the firm level; the significance levels are *10%, **5%, ***1%.

Sources: Registry data and IAB Establishment Panel, 1993–2010. We calculate tasks based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

Table 8: T&O and Career Outcomes for Workers Aged 55 and Over

	(1) Working	(2) Wage growth	(3) Working at the same firm	(4) Manual	(5) Routine	(6) Abstract
Panel A: All						
number of organizational changes	-1.259***	0.031	-0.891***	-0.167	-0.704***	-0.388**
	(0.314)	(0.069)	(0.298)	(0.104)	(0.132)	(0.196)
Panel B: By tasks at baseline						
Manual						
number of organizational changes	-0.917**	0.160*	-0.701	-0.933**	-0.029	0.046**
	(0.442)	(0.089)	(0.443)	(0.429)	(0.044)	(0.022)
Routine						
number of organizational changes	-1.247***	0.052	-1.055***	-0.002	-1.265***	0.021
	(0.338)	(0.067)	(0.355)	(0.020)	(0.325)	(0.021)
Abstract						
number of organizational changes	-1.281***	-0.033	-0.734**	-0.006	-0.003	-1.272***
	(0.327)	(0.096)	(0.298)	(0.006)	(0.010)	(0.327)
Panel C: By education at baseline						
Low						
number of organizational changes	-1.320***	0.113	-0.915**	-0.020	-0.868***	-0.432
	(0.383)	(0.083)	(0.374)	(0.127)	(0.283)	(0.280)
Medium						
number of organizational changes	-1.267***	0.060	-0.953***	-0.251**	-0.748***	-0.267
	(0.325)	(0.058)	(0.318)	(0.125)	(0.139)	(0.183)
High						
number of organizational changes	-1.256***	-0.112	-0.674	0.030	0.005	-1.291***
	(0.397)	(0.169)	(0.421)	(0.032)	(0.062)	(0.393)

Notes: The table reports estimates for the effects of T&O on the career outcomes over a 3-year period of older workers who were employed at a firm in the IAB firm panel and between 55 and 62 at baseline; see equation (3) in the text. Panel A displays results pooled for older workers. Panels B and C break down the analysis by task usage at baseline (manual, routine, abstract) and education (low, medium, high). The main variable of interest in both the 3- and the 6-year regressions, "number of changes," which measures the number of organizational changes in the baseline firm between t-3 and t, ranges between zero and four. We consider the following outcome variables: whether the individual is employed (at any firm in Germany) at time t (column (1)); wage growth (column (2); defined only for workers who are employed at t); whether the worker is employed at the same firm as at baseline at time t (column (3)); and whether the individual performs manual, routine or abstract tasks at time t (columns (4) to (6)). The dependent variables take the values of either 0 or 100 so that the reported coefficients refer to the impact of one additional organizational change on the percentage point change of the particular event (columns (1) and (3) to (6)) or the percent increase in wages (column (2)). All regressions control for age and age squared at baseline, foreign status, year fixed effects, firm size (measured as log employment at baseline), commuting zones (a regional control derived by dividing West Germany into 50 regions based on commuting patterns), and industry (one digit controls). With the exception of the wage growth regressions, they also include pre-estimated fixed firm effects and fixed worker effects calculated using the seven years prior to the baseline year (cf. Card, Heining, and Kline, 2013). The unit of observation is the worker/year, and the full sample (all workers between 55 and 62 at baseline) includes 8,154,560 observations. Standard errors are clustered at the firm level; the significance levels are *10%, **5%, ***1%.

Sources: Registry data and IAB Establishment Panel, 1993–2010. We calculate tasks for the internal reshuffling regression based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

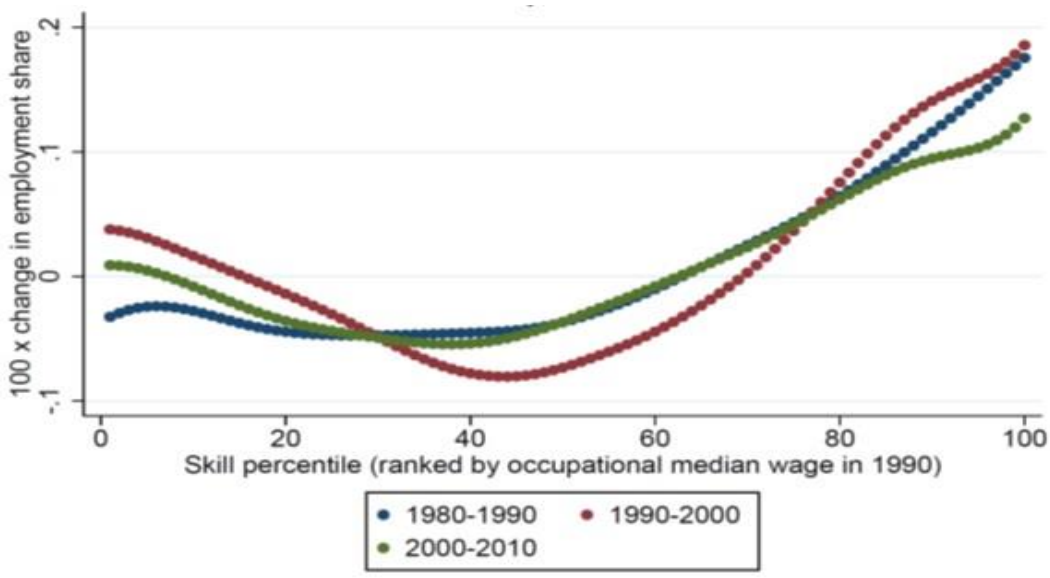
Table 9: T&O, Career Outcomes of Routine Workers, and Labor Market Institutions

	all routine workers				routine workers, 55 and older			
	(1) working	(2) abstract	(3) abstract, same firm	(4) abstract, other firm	(5) working	(6) abstract	(7) abstract, same firm	(8) abstract, other firm
Panel A: by apprenticeship training								
number of changes	0.012	-0.041	-0.042	0.001	-1.377***	-0.03	-0.015	-0.015
	(0.169)	(0.094)	(0.082)	(0.009)	(0.445)	(0.030)	(0.026)	(0.014)
number of changes X share apprentices	-1.274	6.304***	4.641**	1.663*	3.206	1.418**	1.121*	0.297
	(2.328)	(2.427)	(2.172)	(1.001)	(6.119)	(0.682)	(0.631)	(0.280)
mean apprenticeship share				0.044				
Panel B: by firm's union status								
number of changes	0.439	-0.081	-0.103*	0.022	-0.326	-0.121	-0.043	-0.077
	(0.343)	(0.091)	(0.059)	(0.068)	(0.966)	(0.076)	(0.051)	(0.053)
number of changes X union recognition	-0.474	0.280***	0.245***	0.035	-1.162	0.149*	0.077	0.072
	(0.352)	(0.100)	(0.069)	(0.071)	(1.025)	(0.079)	(0.056)	(0.053)
mean union recognition rate				0.845				
Panel C: by share employees on fixed term contracts								
number of changes	-0.265	0.140**	0.100***	0.040	-2.182***	0.015	0.015	0.000
	(0.176)	(0.058)	(0.046)	(0.026)	(0.482)	(0.024)	(0.020)	(0.011)
number of changes X fixed term contracts	5.210***	0.418	0.694	-0.276	12.357***	0.288	0.332	-0.044
	(1.638)	(0.631)	(0.568)	(0.223)	(2.743)	(0.325)	(0.286)	(0.113)
mean share on fixed term contracts				0.078				

Notes: The table investigates how the apprenticeship share at baseline (Panel A), union recognition (measured as the firm's membership in an employer association) at baseline (Panel B), and the share of employees on fixed term contracts at baseline (Panel C) affect T&O's effects on career outcomes of workers who were employed in one of the firms in the IAB firm panel and performed routine tasks at baseline. Columns (1) to (4) display results pooled for all routine workers. Columns (5) to (8) refer to routine workers who were between 55 and 62 at baseline. The main variable of interest, "number of changes," which measures the number of organizational changes in the baseline firm between t-3 and t, ranges between zero and four. This variable is interacted with the share of apprentices in the firm at baseline (Panel A), whether the firm recognized a union at baseline (Panel B), and the share of employees on fixed term contracts at baseline (Panel C). We consider the following four outcome variables: whether the individual is employed (at any firm in Germany) at time t (columns (1) and (5)); whether the individual is employed in an abstract occupation at time t inside or outside the baseline firm (columns (2) and (6)); whether the individual is employed in an abstract occupation at time t at the baseline firm (columns (3) and (7)) or at an outside firm (columns (4) and (8)) at time t. All regressions control, in addition to the apprenticeship share at baseline in Panel A, union recognition in Panel B, and the share of employees on fixed term contracts at baseline in Panel C, for age and age squared at baseline, foreign status, year fixed effects, firm size (measured as log employment at baseline), commuting zones (a regional control derived by dividing West Germany into 50 regions based on commuting patterns), industry (one digit controls), pre-estimated fixed firm effects and fixed worker effects calculated using the seven years prior to the baseline year (cf. Card, Heining, and Kline, 2013). The unit of observation is the worker/year. Standard errors are clustered at the firm level; the significance levels are *10%, **5%, ***1%.

Sources: Registry data and IAB Establishment Panel, 1993–2010. We calculate tasks for the internal reshuffling regression based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

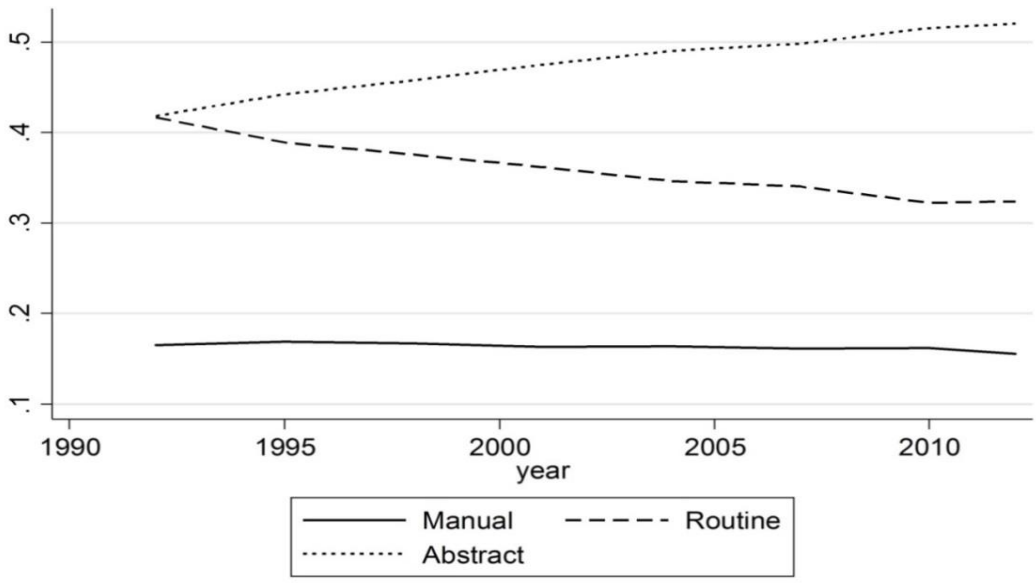
Figure 1a: Changes in Employment Shares by Occupational Skill Percentile



Notes: The figure plots log changes in employment shares by 1990 occupational skill percentile rank using a locally weighted smoothing regression (bandwidth 0.8 with 100 observations), where skill percentiles are measured as the employment-weighted percentile rank of an occupation's median log wage in 1990 (cf. Acemoglu and Autor, 2011). The sample is composed of male workers aged 18-65 in full time employment and representative of the corresponding worker population in West Germany.

Source: 2% random sample of the Registry Data (SIAB7510), 1980–2010.

Figure 1b: Task Shares Over Time

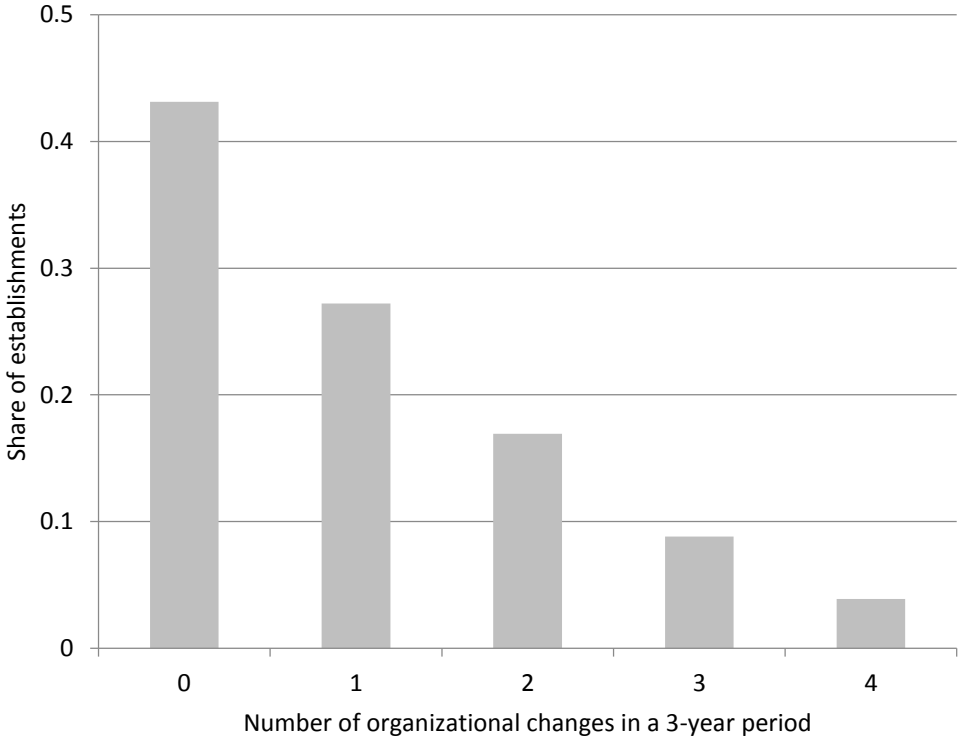


Notes: This figure plots the share of workers in manual-, routine-, and abstract dominated occupations between 1993 and 2012. The unit of observation is the firm/year, and observations are weighted by (baseline) employment and the firm panel weights, to make sure that they are representative of the worker population in West Germany.

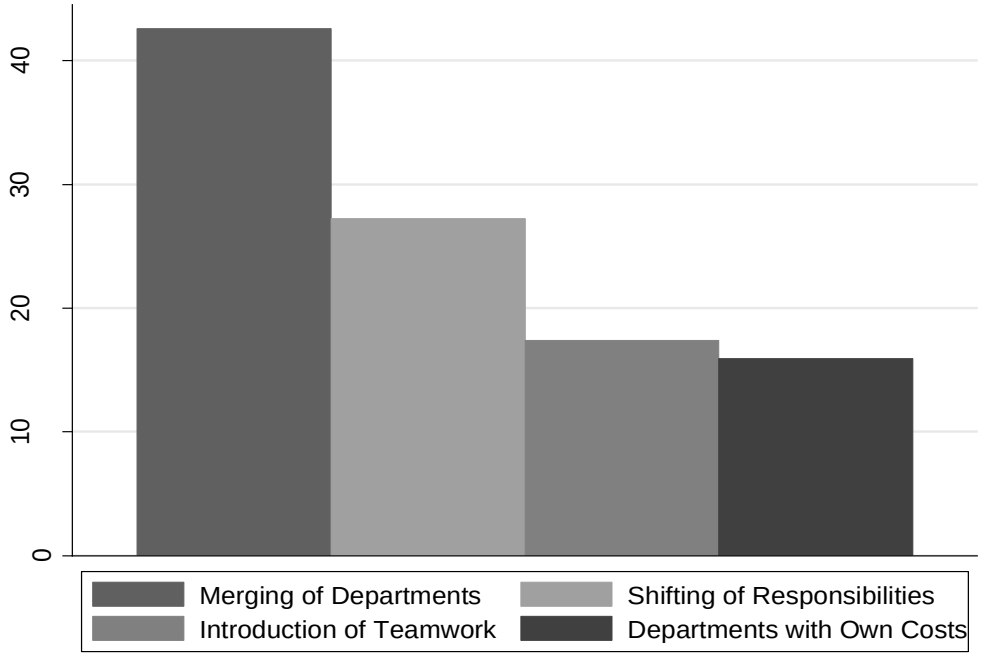
Source: Registry data and IAB Establishment Panel, 1993–2012.

Figure 2: T&O: Descriptives

Panel A: Frequency of T&O



Panel B: Types of T&O

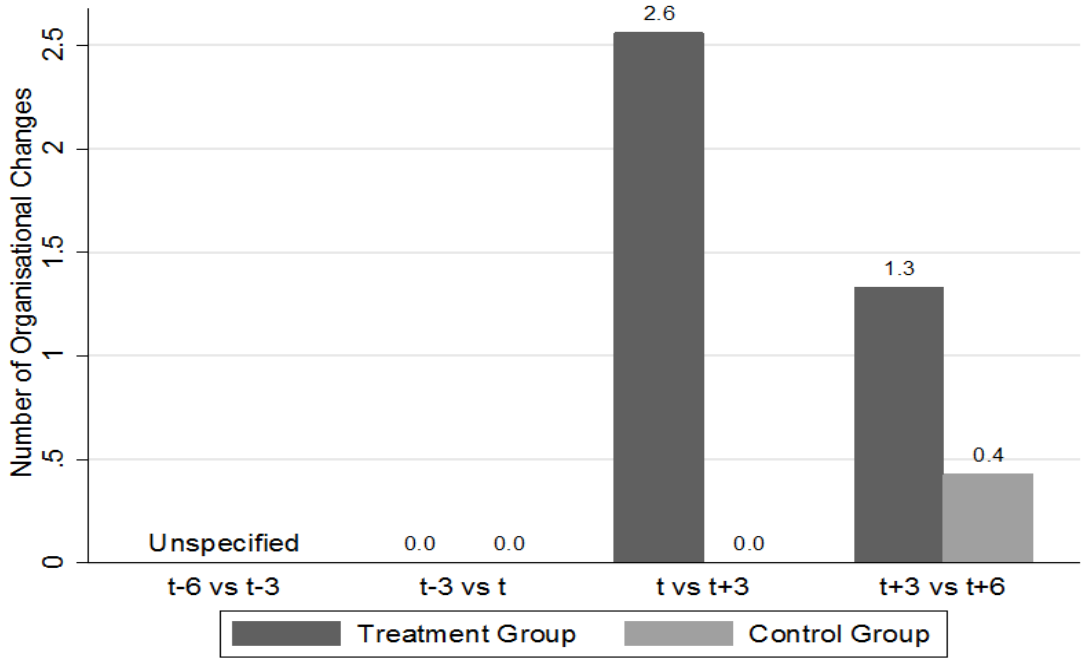


Notes: The figures provide some descriptive evidence about the frequency and type of T&O. Panel A displays the share of firms that implement no, 1, 2, 3, or 4 organizational changes over a 3-year period. Panel B shows the share of firms that over a 3-year period merge or restructure departments internally; transfer responsibilities to subordinates; introduce team work or self-responsible working groups; or introduce profit centers (i.e., units or departments that carry out their own cost and result calculations). The unit of observation is the firm/year, and observations are weighted by (baseline) employment and the firm panel weights, to make sure that they are representative of the worker population in West Germany.

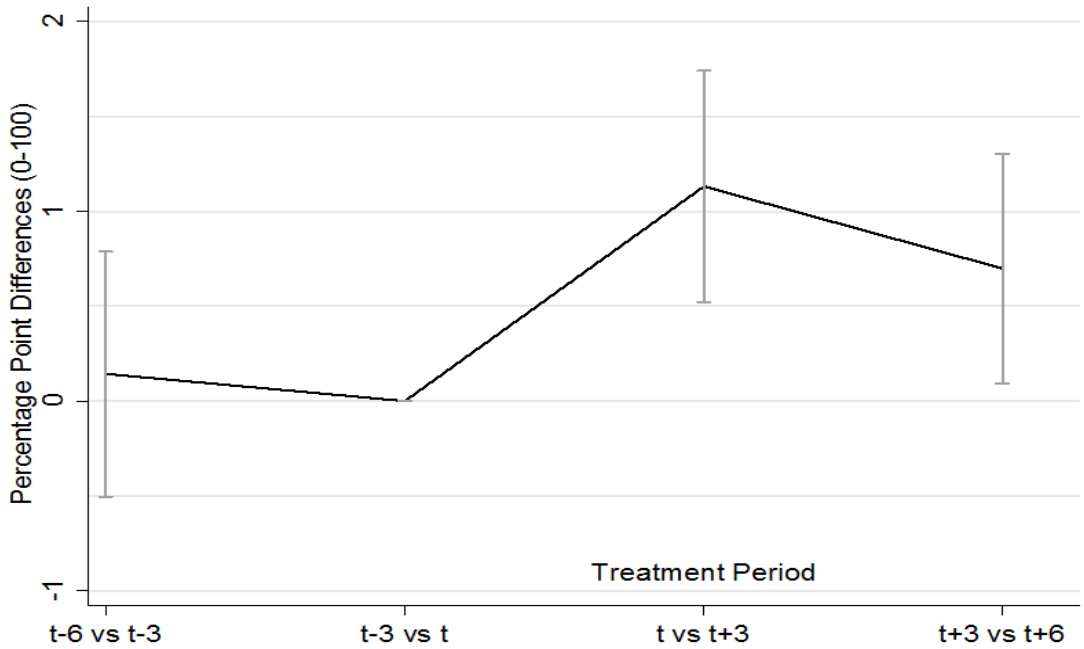
Data Sources: IAB Establishment Panel, 1995-2010.

Figure 3: Routine-to-Abstract Transitions and Employment-to-Non-employment Transitions of Older Workers: Event-Study Analysis

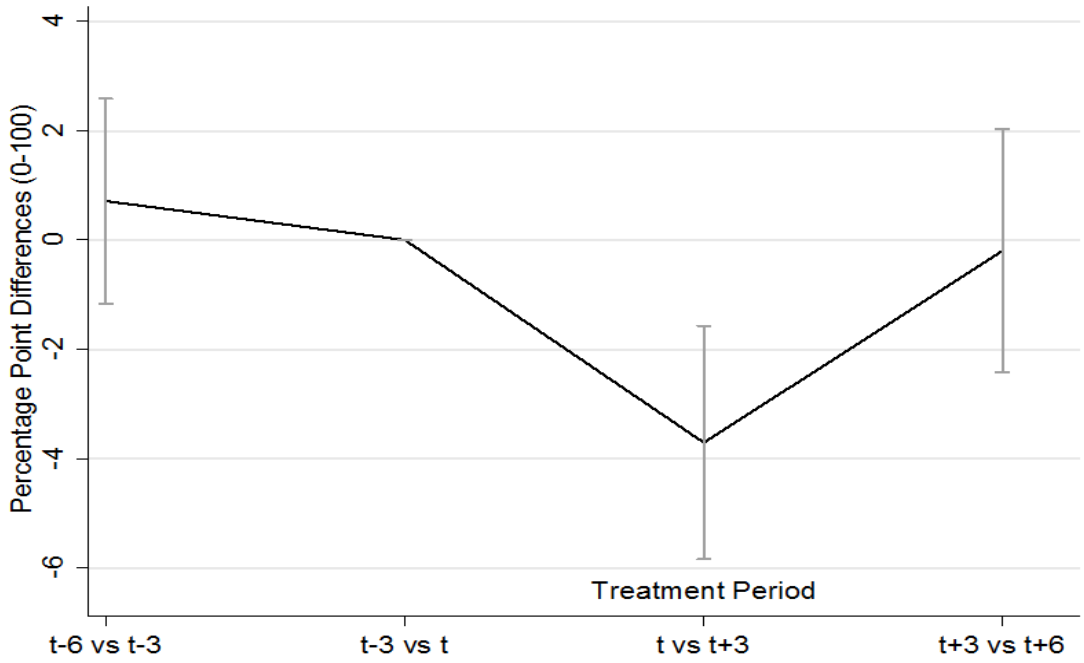
A. Average number of organizational changes, treatment vs. control groups



B. Routine to abstract transitions



C. Employment-to-non-employment transitions of workers aged 55 and older



Notes: The figures display the results of an event-study analysis focused on two outcome variables: upgrades from routine to abstract jobs (Panel B) and transitions from employment to non-employment of workers age 55 and over at baseline (Panel C). "Treatment" firms do not implement any T&O between t-3 and t, but carry out at least two organizational changes between t and t+3 (the "treatment period"). "Control" firms introduce T&O neither between t-3 and t nor between t and t+3. No restrictions on the number of organizational changes implemented by either treatment or control firms between t-6 and t-3 or between t+3 and t+6 are imposed. Panel A contrasts the number of organizational changes that treatment and control firms carry out in each period. Panels B and C then plot the differences in the firm level shares of workers upgrading from routine to abstract tasks (either within or across firms, Panel B) and of older workers who are remain employed at the end of the period (Panel C) between treatment and control firms in each period, where differences between treatment and control firms are normalized to 0 in the t-3 vs t pre-treatment period. In both graphs, firm-level shares vary from 0 to 100. The sample consists of 520 treatment firms and 2626 control firms, which we follow over time. The unit of observation is the firm/year, and observations are weighted by (baseline) employment. The vertical bars in panels B and C indicate 95% confidence intervals.

Sources: Registry data and the IAB Establishment Panel, 1993–2010. We calculate task shares based on a categorization constructed using the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.

Table A1: Classification of Activities into Tasks using BIBB

Manual tasks	Repairing machines Driving vehicles Hosting , serving (e.g. wait tables), accommodating customers Securing, watching over, keeping guard Nursing, personal care of others
Routine tasks	Attending, feeding, equipping machinery Fabricating, manufacturing materials, preparing (e.g., food) Building, constructing and installing appliances Filing, sorting, labeling Billing, computing and bookkeeping[*] Writing and correspondence[**] Buying, selling, managing payments, assisting customers[***]
Abstract tasks	Planning, designing, sketching Executing or interpreting laws, rules, or regulations Analysis and research Computing and programming Educating, training, teaching, consulting Publicizing, presenting, disseminating Hiring, management and control, organizing or coordinating Billing, computing and bookkeeping[*] Writing and correspondence[**] Buying, selling, managing payments, assisting customers[***]

Notes: This table categorizes the activities in the 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey into routine, manual, and abstract tasks. Three task sets—billing, calculating, and bookkeeping; writing and correspondence; and buying or selling—are considered routine if the work process is predefined to the last detail and the tasks are highly repetitive or regularly accomplished using a calculator or accounting machine; a photocopier, filing cards, computer lists, microfilm reader; or a cash register or counter, respectively. Otherwise, these activities are classified as abstract.

Data Source: 1991/1992 wave of the German BIBB/IAB Qualification and Career Survey.